

INPUT

Tell the model how to read the values

01 / THE SAS EXAMPLE

```
data test;
  length char_num $8 char_date $9;
  input char_num $ char_date $;
  num_value = input(char_num, best12.);
  date_value = input(char_date, date9.);
  format date_value date9.;
  datalines;
123 15JAN2023
456 01FEB2023
789 31DEC2023
;
run;
proc print data=test noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Make data types and date interpretation part of the request.

Convert character values into numbers and dates. The nine-character date field preserves the full year. Check calendar dates and numeric values; Python need not copy SAS internal date storage.

EXPECTED RESULT

char_num	char_date	num_value	date_value
"123"	"15JAN2023"	123	2023-01-15
"456"	"01FEB2023"	456	2023-02-01
"789"	"31DEC2023"	789	2023-12-31

date_value is displayed here as YYYY-MM-DD. SAS DATE9. displays the same calendar dates differently. Quotes mark source strings, not numeric outputs.

REPORTED EXPECTED-VALUE MATCHES
Prompt A 3/3 | Prompt B 3/3

Reference values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.
Show code in this response. Do not create or run files, or read existing Python files.

REPORTED TRIAL RESULTS / 21 SEP 2026

All three A runs matched the valid inputs

Run	Values match	Assertions
A1	Yes	None reported
A2	Yes	None reported
A3	Yes	None reported

SHARED REPORTED CONVERSION METHODS

pd.to_numeric + pd.to_datetime
Date parsing used format='%d%b%Y'.
Shared methods do not establish identical full responses.

SOURCES: results-20260921.csv; trial-log-20260921.md; analysis-20260921.md

WHY CHOICES REMAIN OPEN

- 1

Agreement is useful evidence

All six runs reportedly used the same two conversion functions. The three valid rows matched in both groups.
- 2

Representation still matters

A date string can display the right day without being a datetime. Numeric and date types need their own checks.
- 3

Keep the conclusion in scope

Three clean inputs do not test invalid values, missing data or other informats. The report cannot establish future reliability.

ANTI-PATTERN: "MAKE IT LOOK THE SAME"

Request numeric values and datetime values, then check both meaning and type. Preserve the source strings for comparison.

Correct-looking text does not prove a correct data type.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.
Show code in this response. Do not create or run files, or read
existing Python files.

- Recreate char_num and char_date as strings, preserving row order.
- The three inputs are valid; month abbreviations are English.
- num_value: use pd.to_numeric(df['char_num']).
- date_value: use pd.to_datetime(df['char_date'],
  format='%d%b%Y'). Keep this column as datetime values.
- Preserve both source columns and add the two result columns.
- Match calendar dates, not SAS's internal numeric date storage.
- Include assertions for num_value = [123, 456, 789] and for
  date_value formatted as YYYY-MM-DD =
  ['2023-01-15', '2023-02-01', '2023-12-31'].
- Also assert that the result columns have numeric and datetime
  types respectively. Calculate them; do not hard-code results.
- Return one complete script and a brief explanation of the format.
This exercise covers these valid inputs only.
```

WHY THESE DETAILS HELP

- 1 Keep sources and results distinct**
Keep char_num and char_date as strings. Add numeric and datetime columns; preserve the original row order.
- 2 State the date interpretation**
%d%b%Y means day, month abbreviation and year. The prompt specifies English months and calendar-date comparison.
- 3 Ask for value and type checks**
B1-B3 reportedly include both kinds of assertions and pass them. The checks cover these three valid inputs.

REPORTED RESULTS / THREE B RUNS

Run	Values match	Assertions
B1	Yes	Passed: values + types
B2	Yes	Passed: values + types
B3	Yes	Passed: values + types

CHECK IT YOURSELF
Compare every value, then inspect both types.
Use formatted dates for checking; retain datetime data.

One behavior, three implementations

Convert valid numeric and English date text while preserving source strings. Compare calendar dates and result types across languages.

SAS / SOURCE BEHAVIOR

```
%macro transform(in=, out=);
data &out;
  set &in;
  num_value = input(char_num, best12.);
  date_value = input(char_date, date9.);
  format date_value date9.;
run;
%mend;
```

Macro transform supplies source/result names. Standalone reference uses LOCALE=en_US; numeric dates are displayed with DATE9.

PYTHON / PANDAS

```
out = df.copy(deep=True)
out['num_value'] = pd.to_numeric(
    out['char_num'])
out['date_value'] = english_dates(
    out['char_date'])
```

english_dates uses pd.to_datetime with %d%b%Y and temporarily sets LC_TIME=C. Returned date_value stays datetime; out is a copy.

R / BASE

```
out <- df
out$num_value <- as.numeric(out$char_num)
old_locale <- Sys.getlocale('LC_TIME')
on.exit(Sys.setlocale(
  'LC_TIME', old_locale))
changed <- Sys.setlocale('LC_TIME', 'C')
if (changed == '') stop('C unavailable')
out$date_value <- as.Date(
  out$char_date, format = '%d%b%Y')
```

Inside translate, on.exit restores LC_TIME. Numeric values use as.numeric; as.Date returns Date objects. Source text is preserved.

A display is not a type
2023-01-15 can be a string or a calendar value. Test numeric and datetime/Date types as well as the displayed values.

Added valid boundary case
The added edge rows use -0.5, 0012, 29FEB2024 and 01MAR2024. They extend local validation, not the original prompts or trial scores.

SAS INPUT function

pandas.to_datetime

R as.Date

Three checks that earn the pass

Use the same inputs and expected behavior in each language. Check the contract as well as the values.

T1 / Original known-answer fixture

INPUT / CONDITION

Original three numeric/date text rows.

EXPECTED

123, 456, 789; 2023-01-15, 2023-02-01, 2023-12-31.

WHAT THIS CATCHES

Wrong informat, date ordering, result values or hard-coded replacements.

T2 / Added valid leap-day / number case

INPUT / CONDITION

-0.5 / 29FEB2024 and 0012 / 01MAR2024.

EXPECTED

Numeric -0.5 and 12; calendar dates 2024-02-29 and 2024-03-01.

WHAT THIS CATCHES

Integer-only conversion, loss of leap-day correctness or hard-coded fixture results.

T3 / Preserve strings, order and types

INPUT / CONDITION

All CSV rows in reverse order, retaining original source strings.

EXPECTED

Input unchanged; source/order preserved; numeric result and datetime/Date result.

WHAT THIS CATCHES

In-place mutation, sorting, loss of 0012 text, or formatted strings replacing calendar values.

REFERENCE CODE + TESTS: `module-08-input/appendix/`

Run: `python tools/run_appendix_tests.py --modules 08`

New reference code, separate from historical trials. SAS checks await execution; scope is stated in the companion README.